Phylogenetic meta-analysis (MCMCglmm) – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$\begin{aligned} \mathrm{Zr}_i \mid \mu_i,\, \sigma_i &\sim \mathrm{Normal}(\mu_i,\, \sigma_i^2) \\ \mu_i &= \beta_0 + u_{species(i)} \\ \mathbf{u}_{species} &\sim \mathcal{N}(\mathbf{0},\, \sigma^2_{species}\mathbf{A}) \end{aligned}$$

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$\begin{aligned} \mathbf{zr} \mid \boldsymbol{\mu},\, \boldsymbol{\sigma} &\sim \mathcal{N}(\boldsymbol{\mu},\, \mathrm{diag}(\boldsymbol{\sigma}^2)) \\ \boldsymbol{\mu} &= \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \\ \mathbf{u}_{species} &\sim \mathcal{N}(\mathbf{0},\, \sigma^2_{species}\mathbf{A}_{60\times 60}) \end{aligned}$$

3. Worked observation (i = 1)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$\underbrace{\mathrm{Zr}_1}_{\text{observed}=0.166} \;=\; 0.409 \;-0.221 \;-0.0214 = 0.166$$

Predicted $\hat{\mu}_1 = 0.188$*; residual* $\hat{\varepsilon}_1 = -0.0214$.